Alcohol Consumption Trends in the United States

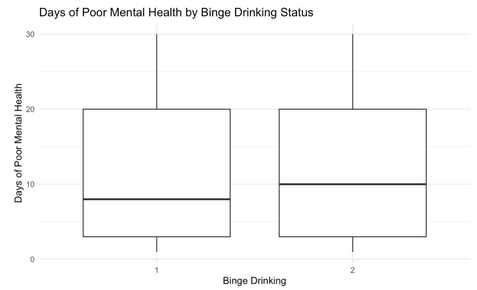
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Goal

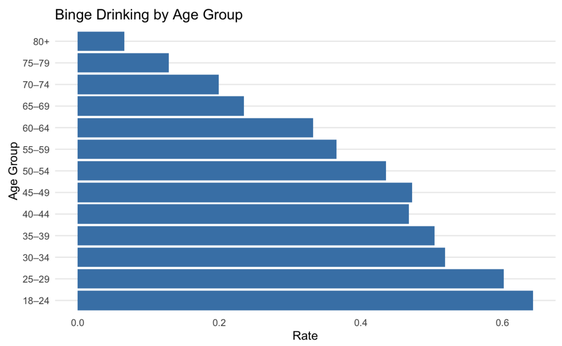
To understand how alcohol consumption behaviors vary among individuals in the United States and how those behaviors are linked to alcohol consumption related health outcomes at the state level.

Introduction

These datasets come from the CDC and contains data on health-related behaviors as well as chronic disease indicators.



Binary Outcome: 1 indicates 'No,' 2 indicates 'Yes'

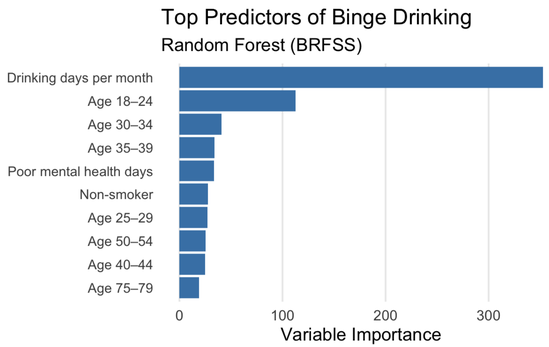


Methodology

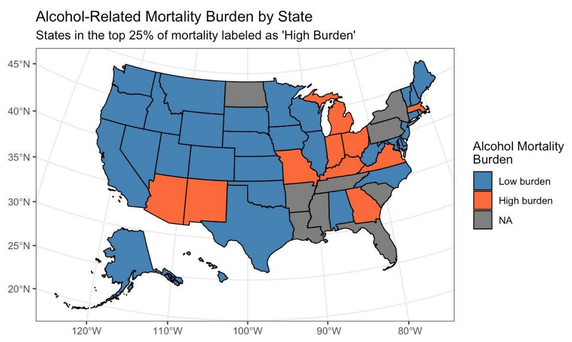
We used Elastic Net + RF because they provide complementary strengths: Elastic Net highlights and identifies the most influential predictors, while RF captures nonlinear patterns and interactions.

Two Modeling Pipelines:

1. Individual-Level
   - Fit Elastic Net + Random Forest
   - **Purpose:** Identify **who** is most at risk
2. State-Level:
   - Aggregated BRFSS into state-level predictors
   - Defined "High Burden" as states in top 25% of liver mortality
   - Fit Elastic Net
   - **Purpose:** Identify population-level factors associated with high burden



Results



Key Predictors of Binge Drinking

- **Age:** Adults (18-34) had the highest binge-drinking rates; those 55+ had much lower risk
- **Mental Health:** More days of poor mental health were linked to a higher likelihood of binge drinking.
- **Socioeconomic Factors:** Lower education and income were linked to slightly higher risk

Population-Level Predictors of Alcohol-Related Mortality

- **Binge Drinking Rates:** States with higher binge drinking showed higher alcohol-related mortality.
- **Income & Education:** Lower-income, lower-education states had greater alcohol-related burden.
- **Regional Differences:** Certain states were consistently high or low burden, indicating regional policy or cultural differences.

Conclusion

Younger adults had much higher odds of binge drinking, while older adults showed lower risk. Once age was included, associations with behaviors like smoking and exercise weakened. State-level results indicated that areas with poorer health and lower income had higher alcohol-related deaths, even after population adjustment. Overall, youth and economic disadvantage were key factors linked to increased alcohol-related harm in the U.S.

Limitations

Phone-based sampling excludes some groups (e.g., those without phones or incarcerated individuals) and may bias results. Self-reported behaviors can be inaccurate due to recall or social desirability bias, and cross-sectional data only show association, not causation. These limitations mean results should be interpreted cautiously to avoid stigmatizing populations or informing harmful policy decisions.

Acknowledgements

We would like to thank the CDC for providing public access to the BRFSS and the US Chronic Disease Indicators datasets, which made our analysis possible.